NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 09

Testing, Deployment and Handover

Test strategy and verified results, deployment summary, operational handover, known limitations and runbooks.

Document control

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Intended readers	Incoming maintainers and operators; assessor checking test evidence
Evidence baseline	Branch New-Changes: commit 716bedb plus the uncommitted Part 3 working-tree changes of 2026-11-05 (v1.0 analysed 3a3f275)
Prepared by	AI-assisted (Claude Code) from repository evidence; team review pending
Approved by	Unknown No approval recorded

Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

Contents

1. 1 Test strategy
2. 2 Verified results
    1. 2.1 Coverage gaps
3. 3 Deployment summary
4. 4 Operational handover
5. 5 Known limitations
6. 6 Runbooks

1 Test strategy

Test levels in the repository

Level	Tooling	What it covers	Runs in CI
Backend unit/integration	Jest + Supertest against the in-process Express app	Validation, auth, rate limiting, error shape, health, body limit, concurrency, sync replay, ownership and conflicts	Yes (CI-01, CI-02)
Android unit (JVM)	JUnit 4, MockK, coroutines-test, shared fakes	ViewModels (auth, dashboard, projects, settings), `SyncManager`, `HabitScoring`	Yes (`./gradlew test`)
Android instrumented	AndroidX JUnit, Compose UI test	Room offline persistence and ownership, splash with reduced motion, screenshot-capture wait conditions	No; needs an emulator or device
Container smoke	Docker + curl `/health`	Image builds and starts in production mode	Backend CI only; never succeeded (CI-02)
Manual acceptance	`docs/ONLINE_OFFLINE.md` six-step check; `docs/DEMO_SCRIPT.md`	Online/offline round trip, account switch, conflicts	No. No recorded results in the repository **Unknown**

2 Verified results

> **v2.0 results (2026-11-05, after the Part 3 changes):** backend **29/29** (adds `part3.test.js`, 13 tests: SSO, push, device tokens, data-file safety, email case). Android unit **42/42** (adds `AuthErrorMessagesTest` 5, `ReminderPlannerTest` 2, `CsvRowTest` 2, one `HabitScoringTest` case). Instrumented **16/16** on an Android 7.0 emulator with the API stopped (AUD-22 explains the 14/16 run with a live API). The on-device walkthrough covered the sign-in error, all screens, three languages, the offline round-trip, the reminder and test notifications, and the Settings crash fix. The tables below are the v1.0 baseline results.

v1.0 run on 2026-11-05 against commit 3a3f275, before the Part 3 changes.

Backend: `cd backend; npm test` **, 2 suites, 16 tests, 16 passed, 31.7 s**

ID	Suite	Tests	Result
TST-01	`tests/server.test.js` : JWT secret refusal; concurrent updates; Zod validation (register, task enum/date, strict profile); error shape and production masking; expired token 401; 429 after 10 logins; `/health` ; 413 over 1 MB	11	Passed
TST-02	`tests/sync.test.js` : ordered replay with cascade delete; lost response plus restart does not duplicate or resurrect; cross-account project rejected; invalid payload 400 and no-auth 401; edit of deleted item gives 409	5	Passed

Android: `./gradlew :app:testDebugUnitTest --rerun` **, 7 classes, 32 tests, 0 failures, BUILD SUCCESSFUL**

ID	Class	Tests	Result
TST-03	`AuthViewModelTest`	9	Passed
TST-04	`DashboardViewModelTest`	7	Passed

ID	Class	Tests	Result
TST-05	`SyncManagerTest` : acknowledged operations removed in sequence; network failure keeps pending changes; pending writes block refresh overwrite; offline reads use cache while auth errors surface	4	Passed
TST-06	`HabitScoringTest`	4	Passed
TST-07	`SettingsViewModelTest`	4	Passed
TST-08	`ProjectListViewModelTest`	3	Passed
TST-09	`ExampleUnitTest` (template)	1	Passed

Not executed for this package

ID	Test	Count	Status
TST-10	Instrumented: `OfflinePersistenceTest` (1), `SplashIntroScreenTest` (4), `ScreenshotCaptureTest` (10), `ExampleInstrumentedTest` (1)	16	Not run at v1.0; run at v2.0: 16/16 passed on an emulator
TST-11	Docker build and smoke test	1	Not run; Docker daemon not running Unknown
TST-12	Manual acceptance check (`ONLINE_OFFLINE.md`)	6 steps	v2.0: steps 1–4 exercised on the emulator (offline create, then reconnect, then present on the server, no duplicate); steps 5–6 (account switch, cross-client delete conflict) not run Unknown

2.1 Coverage gaps

- No unit tests for `ProjectDetailViewModel` , `TaskListViewModel` , `HabitsViewModel` , `RemoteConfigViewModel` , or the repositories' Room paths Implemented .
- No backend tests for habits CRUD, password change, account deletion or the localisation/config endpoints Implemented .
- No Room migration tests (AUD-12); no tests for the defects in AUD-01, AUD-11, AUD-17 and AUD-18.
- A Kover report is configured but no coverage figure was produced or recorded Unknown .

3 Deployment summary

Full procedures are in NX-11-DEP; release gating is in REL-CHECK. In short: run the API with a persistent `JWT_SECRET` and persistent storage (Node host or `backend/compose.yaml`), put it behind HTTPS, and build the app with `-PNEXUS_API_URL=https://…/api/` . No environment beyond local development exists today (CON-02).

4 Operational handover

Handover register

Item	Current state
Source repository	`github.com/jBlack-MC/Nexus-app` ; default branch `main` ; working branch `New-Changes` Implemented
Repository owner / admins	Owner account `jBlack-MC` ; other collaborators Unknown

Item	Current state
Secrets in use	`JWT_SECRET`; since v2.0 also `GOOGLE_CLIENT_ID` (not secret), `FIREBASE_SERVICE_ACCOUNT_BASE64` (secret), and the upload keystore at `%USERPROFILE%\.nexus\nexus-upload.jks` with its password in git-ignored `keystore.properties` on the development PC. **Back these up.** CI uses a hard-coded test-only JWT value Implemented
CI variables	`NEXUS_API_URL` (enables release artifacts). Whether it is set: Unknown. Release jobs were skipped in the retrieved runs Inferred
Hosted services	None Implemented
Monitoring and alerting	`/health` endpoint and Docker `HEALTHCHECK`. No external monitoring Implemented
Backups	None configured Implemented. See RB-04
Open pull requests	#27 Dependabot Zod 4. Do not merge until IMP-02 is done

5 Known limitations

- Single-instance JSON store (CON-01); not suited to concurrent hosting or large data.
- No background sync while the app process is stopped (CON-03).
- Last-upload-wins for edits; no field-level merge (FR-17).
- Account recovery and collaboration are not implemented (NX-01-PRD scope). Google SSO and FCM push need the team's Google Cloud and Firebase projects before they work end to end.
- isiZulu and Setswana translations are AI-drafted and need first-language review (FR-18).
- Open defects: see NX-08-AUD.

6 Runbooks

Operational runbooks

ID	Situation	Steps
RB-01	API exits at start with "FATAL: JWT_SECRET is missing, too short..."	Generate a secret: `node -e "console.log(require('crypto').randomBytes(48).toString('base64url'))"`. Set `JWT_SECRET` in the host's secret store (not in git) and restart. Reuse the same value on later restarts.
RB-02	Health check	`curl -s https://<host>/health` should return `{"status":"ok","uptime":…}`. Locally: `http://localhost:5263/health`.
RB-03	All users suddenly see empty data after a restart	Likely AUD-01. **Stop the API immediately** so no further writes occur. Inspect logs for "Failed to load database". Restore the data file from the latest backup (RB-04) and restart. Without a backup, the data overwritten by the empty store cannot be recovered.
RB-04	Back up and restore data Recommendation	Back up: stop the API, or accept a crash-consistent copy (writes are atomic renames), then copy `nexus-data.json` (container: `docker compose cp api:/data/nexus-data.json ./backup-YYYYMMDD.json`). Restore: stop the API, replace the file, start the API, check RB-02. Treat backups as personal data.

ID	Situation	Steps
RB-05	Users cannot sign in (HTTP 429)	The limit is 10 auth requests per 15 minutes per IP, held in memory. Wait 15 minutes or restart the API. Behind a proxy, check AUD-06.
RB-06	Rotate `JWT_SECRET`	Set the new value and restart. Every existing session becomes invalid; users see "Sign in again to sync saved changes", and their queued changes upload after they sign in.
RB-07	A user reports the sync banner stuck on "review required"	Ask them to tap Review. "Discard item edits" removes only that item's queued edits, then the rest uploads. If they choose "Keep changes", nothing uploads until the conflict is resolved (AUD-19).
RB-08	CI runs fail within seconds	Open the run with `gh run view <id>` and read the annotations. If they show "job was not started", the cause is an account or billing restriction, not the code (AUD-04).

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